# ASE 230 Tools - GitHub in a Web Browser

# Learning Goals

Using only a web browser, you will be able to:

1. Create and verify a GitHub account.
2. Open a repository and upload course files.
3. Find a file and download it.
4. Download an entire repository as a ZIP file.

**You do not need to know Git commands for this lesson.**

# What Is GitHub?

GitHub is a website that stores and shares projects.

- A **repository** (or **repo**) is a project folder on GitHub.
- A repository can contain source code, documents, images, and subfolders.
- A **commit** is a saved set of changes with a short description.

```
GitHub account
    └── repository
```

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# Keep Private Information Out of GitHub

Before uploading, check every file.

**Never upload:**

- Passwords
- API keys or access tokens
- Private keys
- Personal or confidential information

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# 1. Create a GitHub Account

Open this page in a web browser:

https://github.com/signup

Follow the instructions to create a free personal account.

- Use an email address that you can open now.
- Choose a username you can recognize and share with the instructor.

# Verify Your Email Address

GitHub sends a verification message to your email address.

1. Open the message from GitHub.
2. Click the verification link.
3. Return to GitHub and sign in.

If you cannot find the message:

```
Profile picture → Settings → Emails → Resend verification email
```



Some GitHub features are unavailable until your

# Protect Your Account

GitHub strongly recommends two-factor authentication (**2FA**).

```
Profile picture → Settings
→ Password and authentication
→ Enable two-factor authentication
```

Save your recovery codes in a safe place—not inside your course repository.

## 2. Before You Upload

You need a repository where you have permission to add files.

For a course assignment, this may be:

- A repository created for you by GitHub Classroom
- A repository created by the instructor
- Your own repository

Open the repository in your browser and confirm

# Understand the Repository Page

Near the top of the page, GitHub shows the repository owner and name:

```
owner / repository
```

The file list below it shows the current folder.

- Click a folder name to open it.
- Click a file name to view it.
- Use the breadcrumb path above the file list to

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# Upload Files in the Browser

1. Open the correct repository and destination folder.
2. Select **Add file**.
3. Select **Upload files**.
4. Drag files or a folder onto the page, or select **choose your files**.
5. Wait until every filename appears in the upload list.

# Check the Folder Structure

Make sure you did not create an extra nested folder.

```
Correct
assignment1/
├── index.php
└── README.md
```

```
Wrong
assignment1/
└── assignment1/
    ├── index.php
    └── README.md
```

# Save the Upload as a Commit

At the bottom of the upload page:

1. Enter a short commit message, such as:

```
Submit assignment 1
```

2. Follow the repository's instructions to commit directly or propose the changes on a new branch.
3. Select **Commit changes** or **Propose changes**.



The exact button depends on the repository's

# Confirm That the Upload Worked

Do not stop when the upload screen closes.

1. Return to the repository's file list.
2. Open the assignment folder.
3. Confirm that every required file is present.
4. Click each important source file and check its contents.
5. Refresh the page once and confirm the files

# Uploading a Revised File

To submit a corrected version:

1. Return to the same repository folder.
2. Select **Add file → Upload files**.
3. Upload the revised file with the **same filename**.
4. Enter a message such as `Fix assignment 1`.
5. Commit the change.

Check the displayed file afterward to make sure the intended version is online.

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# Browser Upload Limitations

Browser upload is convenient for small assignments, but it is not full Git synchronization.

- It does not automatically know which local files changed.
- A deleted local file is not automatically deleted from GitHub.
- Uploading to the wrong folder creates the wrong repository structure.

# 3. Find Course Files

Open the ASE 230 repository:

https://github.com/nkuase/ase230

To browse normally:

1. Start at the repository's main page.
2. Click a module or folder.
3. Continue clicking folders until you see the required file.
4. Use the breadcrumb path to go back.

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# Find a File by Name

From a repository page, use GitHub's **Go to file** control to search the repository's filenames.

1. Select **Go to file** above the file list.
2. Type part of the filename.
3. Select the matching result.

You can also press `t` while viewing the repository to open the file finder.

# View and Copy a Source File

When you open a text or source file, GitHub displays its contents.

- Use the copy button to copy the file's contents.
- Select **Raw** to view only the original file contents.
- Browser search (`Ctrl+F` on Windows/Linux or `Cmd+F` on macOS) finds text on the current page.



Copying displayed code is useful for a small

# Download One File

For a text or source file:

1. Open the file on GitHub.
2. Select **Raw**.
3. Save the page from your browser:
   - Windows/Linux: `Ctrl+S`
   - macOS: `Cmd+S`
4. Keep the original filename and extension.

After saving, open the file in VS Code and confirm

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# 4. Download the Entire Repository

From the repository's main page:

1. Select the green **Code** button.
2. Select **Download ZIP**.
3. Open your Downloads folder.
4. Extract or unzip the downloaded file.
5. Open the extracted folder in VS Code.

# File Download vs. ZIP Download

| What you need | Recommended action |
|---|---|
| One small source file | Open it → **Raw** → Save |
| Several related files | **Code → Download ZIP** |
| The complete course repository | **Code → Download ZIP** |

# Common Problems

| Problem | What to check |
| --- | --- |
| No **Add file** button | Sign in and confirm you have write permission. |
| Upload is rejected | Check file size, repository rules, and secret warnings. |
| Instructor cannot find the file | Check repository, branch, and folder location. |
| Downloaded file opens as HTML | Download through **Raw**, not the normal file page. |

# Final Checkpoint

You should now be able to:

1. Sign in with a verified GitHub account.
2. Explain the difference between an account, repository, folder, file, and commit.
3. Upload files using **Add file → Upload files**.
4. Confirm the files after committing them.
5. Find a file by browsing folders or using **Go to file**.
6. Download one file or an entire repository ZIP